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ORIGINAL ARTICLE

SPORT INJURIES AND REHABILITATION

Shoulder injury patterns and trends in the national football league: an epidemiological analysis

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ABSTRACT

BACKGROUND: Athletic movements in American football render the shoulder vulnerable to injury. The aim of this study was to explore the epidemiology of shoulder injuries in the National Football League (NFL).

METHODS: Injury data from the 2019-2022 seasons were extracted from official NFL website and secondary resources. Injury parameters included type, severity, week, position and other demographic characteristics. Injury rates were calculated per 1000 athletic exposures (AE) and statistical analyses were used to determine whether any significant differences exist in injury patterns across different positions and types.

RESULTS: A total of 6732 injuries were reported during our study period, of which 452(6.7%) were attributed to the shoulder, leading to an injury rate of 4.5 per1000 AE. Linebackers, cornerbacks, wide receivers, and safety positions had significantly higher numbers of shoulder injuries than other defensive and offensive positions. Most injuries were mild (66%, $P<0.001$). Muscular (40.9%) and ligamentous (37.4%) injuries were the most common injury types. Fractures and ligamentous injuries were significantly more severe than muscular and other injuries ($P<0.05$). Even though week 17 had the highest mean number of shoulder injuries at 10.75/season, there was no apparent shoulder injury distribution trend across the football season.

CONCLUSIONS: Shoulder injuries are common in American football. Introducing a deidentified injury registry that provides detailed injury information in the NFL may help develop better prevention strategies and protocols.

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KEY WORDS: Sprains and strains; Sports; Football.

American football is a popular contact sport that is known for its intensity and high injury risk, with injury reports reaching 20.9 per 1000 athletic exposures (AE) in high school levels, and 44.2 per 1000 AE in college levels.[1, 2] The National Football League (NFL) is the primary professional football organization of the United States, and serves as the main ambassador for the sport worldwide. Some studies have observed that up to 81% of football players experience an injury each year, which is the highest rate of reportable injuries compared to other organized sports in the United States.[3] Football plays expose athletes to repetitive collisions and violent tackles, which can lead to numerous injuries involving the head, upper extremities, chest, spine, abdomen, and lower extremities. In particular, the risk of traumatic head injuries is significant, with one of the highest documented rates of concussions and neurological disorders in sports, including recent links to chronic traumatic encephalopathy (CTE) diagnosed *post-mortem*.[4] However, fewer studies have examined football injuries that involve the upper extremity, in particular the shoulder girdle. Shoulder injuries can occur via falls, tackles, collisions or through repetitive/overuse maneuvers like blocking or overhead throwing, which can cause harm to the joint from the extremes of range of motion or high force/velocity, and lead to significant pain and debilitation.[5-8] Most prior work involving

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shoulder injuries in American football focused on isolated injuries only, frequently instability or fractures, and there is a paucity of literature that addresses and analyzes the epidemiology of all shoulder-related injuries.[9, 10] There has recently been a renewed focus on player safety in American football at all levels through both updated equipment and major rule changes, especially since 2017, increasing the need for an updated understanding of injury patterns over time.[11] The purpose of this study is to explore patterns and trends of shoulder injuries in the NFL over time, including location, severity, position, and games missed to help understand risk factors and optimize training regimens for prevention.

Materials and methods

Study design

A descriptive epidemiological study of shoulder injuries in the NFL was conducted, with player injuries extracted from the injury report database on the official NFL website.[12] Data extraction was conducted on June 2023 by three authors. All injuries listed from the beginning of the 2019 season and until the end of 2022 season were included in the present study. Additional information regarding the type of shoulder of injuries was obtained from media reports and websites.[13-15] These websites compile, and format injury information released by the NFL and are considered trustworthy sources of information. Although no comprehensive source for player injuries exists, other prior epidemiology studies have used similar methodology (utilizing data obtained from media reports and publicly accessible websites) for compiling information.[16-18]

Definitions

American football, including the NFL, involves several player positions split between offense, defense, and special teams. Offensive positions include the Center, Quarterback, Offensive Linemen, Running Back, Fullback, Tight End, and Wide Receiver. Defensive positions include the Defensive Linemen, Linebacker, Cornerback, and Safety. Special team positions include the Kicker, Punter, Kick Returner, Punt Returner, and Long Snapper (with associated offensive and defensive linemen and blockers). Shoulder injuries were included if recorded on the weekly injury report published on the official NFL website.[12] Shoulder injuries that were cited for consecutive weeks for the same player on the same side were considered as a single injury. Injuries where multiple anatomic locations were cited in the same week were considered multiple injuries. For the scope of this study, shoulder *sprains* were defined as injuries occurring when the ligaments of the shoulder were torn or overstretched, and a shoulder *strain* as an injury occurring when the muscles or tendons of the shoulder were torn or overstretched. Accordingly, injuries were categorized as follows: Injuries reported as "labral tears," "shoulder subluxations" or "shoulder dislocations," "shoulder sprain," "AC joint sprain," "acromioclavicular separation," or "sternoclavicular dislocation" were grouped as "ligamentous" injuries. Those that were reported as "rotator cuff tear," "partial rotator cuff tear" or "shoulder strain" were categorized "muscular/tendinous" injuries. Injuries reported as fractures of the shoulder or scapula were grouped as "fractures." Shoulder injuries that did not adhere to any of the above categories were labeled "other." Injury timing was reported according to the week attributed to the injury. For the scope of this study, playoff games were not included to avoid underreporting injury rates given the smaller number of teams involved. The 2020-2021 season was the final season to include a 16-game schedule, while subsequent seasons were expanded to 17 games. As such, when reporting the mean number of injuries per week, we divided the total number of injuries sustained for every week by the number of seasons that included that week. For analysis, the season was divided into three parts: early season (weeks 1-6), midseason (weeks 7-12) and late season (weeks 13 – onwards), with rates describing the 2021-2022 and 2022-2023 seasons weighted to account for this extra game. Injuries were considered mild if the number of games missed was less than or equal to one game, and moderate if between 2 and 6 games (inclusive). They were considered severe if the number of games missed exceeded 6 games. This was categorized after reviewing relevant literature exploring injuries in NFL and other sports, and consulting expert shoulder and sports surgeons.[19-21]

Data processing

The final dataset included information on player position, injury location, injury type, injury year, week of injury during the season, and number of games missed. Information was also included on player demographics, including age at injury, height, weight, Body Mass Index (BMI), and previous playing experience (years). To calculate injury rate (IR), a methodology similar to previous studies was employed.[16] Athletic exposures were calculated according to a "one game per athlete" rule: the number of games was multiplied by the number of teams and the number of

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active players on each team to calculate the athletic exposures for each season. Injury rates were reported as the number of injuries per 1000 AE.

Statistical analysis

Descriptive statistics were used to report demographic characteristics of injured players. To determine whether significant differences existed between different offensive and defensive positions, a One-Way Analysis of Variance (ANOVA) Test was used. In the case of significance with ANOVA testing, the Tukey's Test or Games Howell Test was employed for post hoc testing depending on whether the assumption of homogeneity of variances was violated. An independent *t*-test was used to compare shoulder injuries among specialty team positions, since only two such positions were reported in our study. When statistically, comparing number of injuries across all positions, an ANOVA Test was used, with specialty teams excluded due to the significantly lower number of plays played compared to defensive and offensive positions. To determine whether significant differences existed between the number of shoulder injuries according to type, a Kruskal-Wallis Test was used, and the Bonferroni correction was implemented to report the adjusted P value and determine significance. P values less than 0.05 were considered statistically significant. The Statistical Package for the Social Sciences for Windows software (Microsoft Corp., Redmond, WA, USA) version 25.0 (IBM SPSS, 2017; IBM Corp., Armonk, NY, USA) was used for all statistical analyses.

Results

Injury rates and demographics

During the study period, a total of 6732 injuries across all anatomic locations were recorded in the NFL, comprising an injury rate of 67.1 per 1000 AE. Total injury rates were consistent throughout the studied period, but 2020 had the highest rate at 71.3 per 1000 AE, while 2021 had the lowest rate at 65.2 per 1000 AE (Table I). Out of these 6732 injuries, a total of 452 (6.7%) in 360 NFL players were attributed to the shoulder, leading to an injury rate of 4.5 per 1000 AE. The average NFL player who sustained a shoulder injury in this study was 26 years of age at the time of injury, had a height of 188.1 cm (range: 170-203 cm), a weight of 110.5 kg (range: 76.5-164.25 kg), and BMI of 31.1 (range: 23.6-47.7). In addition, the average experience of a player sustaining a shoulder injury in the NFL was found to be around 6 years (range: 0-15 years). The number of shoulder injuries was highest in 2020 at 130 and lowest in 2022 with 81 injuries, despite the increased number of games in later years, leading to an injury rate of 5.35 per 1000 AE in 2019 and 3.1 per 1000 AE in 2022 (Table I).

Table I.—*Distribution of total injuries, shoulder injuries, and injury rates during our study period (2019-2022).*

Year	AE	Total injuries	Total injury rate	Shoulder injuries	Shoulder injury rate (per 1000 AE)
2019	23.552	1573	66.788	126	5.35
2020	24.576	1752	71.289	130	5.3
2021	26.112	1702	65.181	115	4.4
2022	26.112	1705	65.296	81	3.1
Total	100.352	6732	67.084	452	4.5

AE: athletic exposures.

Shoulder injury and player position

Between player positions, 258 injuries (57%) occurred in defensive players, 192 injuries (42.5%) in offensive players, and only 2 injuries (0.44%) occurred in special team players. There was no significant difference in the number of shoulder injuries between offensive and defensive positions (P=0.371). That being said, shoulder injury rates among special team positions were significantly less than that of offensive (P=0.006) and defensive positions (P=0.014). The positions with the highest number of sustained shoulder injuries were linebackers and cornerbacks with 66 shoulder injuries (15%) each, followed by Wide

Table II.—*Distribution of shoulder injuries according to playing position (2019-2022).*

Position	2019	2020	2021	2022	Total	Percentage
Defense						
Linebacker	17	17	26	6	66	15
Cornerback	18	15	23	10	66	15
Safety	11	14	10	10	45	10
Defensive tackle	7	17	10	6	40	9
Defensive end	12	13	7	5	37	8
Offense						
Wide receiver	19	13	14	10	56	12
Running back	13	5	9	8	35	8
Guard	12	13	3	7	35	8
Offensive tackle	5	8	4	8	25	6
Tight end	6	6	4	5	21	5
Quarterback	4	6	2	5	17	4
Center	2	1	3	0	6	1
Fullback	0	0	1	0	1	0
Specialty						
Kicker	0	1	0	0	1	0
Long snapper	0	0	0	1	1	0
Total	126	129	116	81	452	100

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TABLE III.—*Significant statistical comparisons between different offensive and defensive playing positions.*

Significant statistical comparisons	Mean difference	P value
Linebacker > Tight end	11.25	0.011
Linebacker > Offensive tackle	10.25	0.03
Linebacker > Quarterback	12.25	0.004
Linebacker > Fullback	16.25	<0.001
Linebacker > Center	15	<0.001
Cornerback > Tight end	11.25	0.011
Cornerback > Offensive tackle	10.25	0.03
Cornerback > Quarterback	12.25	0.004
Cornerback > Fullback	16.25	<0.001
Cornerback > Center	15	<0.001
Wide receiver > Quarterback	9.75	0.047
Wide receiver > Fullback	13.75	0.001
Wide receiver > Center	12.5	0.003
Safety > Tight end	6	0.048
Safety > Quarterback	7	0.03
Safety > Fullback	11	0.009
Safety > Center	9.75	0.005

Receivers with 56 injuries (12%) (Table II). Among offensive positions, Wide Receivers had a higher number of injuries when compared to centers (P=0.031) and fullbacks (P=0.03). Similarly, tight ends had a significantly higher number of injuries when compared centers (P=0.042) and fullbacks (P=0.004). There was no significant difference in number of injuries between the different defensive positions (P=0.183) and between the different specialty positions (P=1.00). When exploring statistical differences in number of shoulder injuries between offensive positions and defensive positions, linebackers, cornerbacks, wide receivers, and safety positions had higher numbers of shoulder injuries, with significant statistical comparisons shown in Table III.

Shoulder injury severity

The average number of games missed following a shoulder injury in our study was 1.68 (~2 games), with a range between 0 and 16 games (Table IV). Mild injuries comprised most injuries (N.=300, 66%) and led to an average of 0.4 games missed, with 190 of these injuries not leading to any missed games. A total of 126 injuries (28%) were classified as moderate, resulting in an average of 2.8 (~3) missed games, and 26 injuries (5.8%) were classified as severe, resulting in an average of 11 missed games (Table IV). Mild injuries were significantly greater in number than moderate injuries (P<0.001) and severe injuries (P<0.001). The number of moderate injuries was also significantly greater than that of severe injuries (P=0.012) (Table IV).

TABLE IV.—*Distribution of shoulder injuries in our study by severity (2019-2022).*

Year	Mild (%)	Moderate (%)	Severe (%)	Total (%)
2019	74 (58.7)	37 (29.4)	15 (10.9)	126 (27.9)
2020	92 (70.2)	34 (26)	5 (3.8)	131 (29.0)
2021	75 (65.8)	34 (29.8)	4.4	114 (25.2)
2022	59 (72.8)	21 (25.9)	1 (1.2)	81 (17.9)
Total	300 (66.3)	126 (27.9)	26 (5.8)	452 (100)

Shoulder injury types

When stratifying shoulder injuries according to type, 185 injuries (40.9%) were reported to be muscular/tendinous in nature, leading to an injury rate of 1.84 per 1000 AE. Ligamentous injuries were the next most prevalent, with 169 injuries (37.4%) and injury rate of 1.68 per 1000 AE. Only 6 injuries (1.3%) were reported to be fractures, resulting in an injury rate of 0.06 per 1000 AE, and 92 injuries (20.4%) had injury types categorized as "other," with an injury rate of 0.92 per 1000 AE (Table V). Non-parametric comparison testing revealed no statistically significant differences between different injury types after adjusting significance values by Bonferroni correction. Fractures overall resulted in the most severe injuries, contributing to an average of 3.33 missed games. Ligamentous injuries followed with an average of 2.49 missed games, and muscular/tendinous injuries had an average of 1.25 games missed. Unclassified "other" injuries had the lowest severity with only 0.98 games missed on average. These differences compared to fractures were significantly more severe than muscular/tendinous injuries (P=0.005) and other injuries (P=0.003). Similarly, ligamentous injuries were also significantly more severe than muscular/tendinous injuries (P=0.005)

TABLE V.—*Comparison of shoulder injuries by type, incidence rate, and average games missed.*

Shoulder injury type	N.	Incidence rate (per 1000 AE)	Average games missed (median)
Fractures	6	0.06	3.33 (3)
Ligamentous	169	1.68	2.49 (1)
Muscular/tendinous	185	1.84	1.25 (1)
Other	92	0.92	0.98 (1)
Total	452	4.5	1.68 (1)
Comparisons of severity	**Test statistic**	**Standard error**	**Adjusted significance**
Muscular/tendinous – ligamentous	44.52	13.3	0.005
Muscular/tendinous – fractures	165.65	51.83	0.008
Ligamentous – fractures	121.13	51.9	0.118
Other – muscular tendinous	18.161	15.94	1
Other – ligamentous	62.683	16.19	0.001
Other – fractures	183.81	52.65	0.003

P values <0.05 shows significance.

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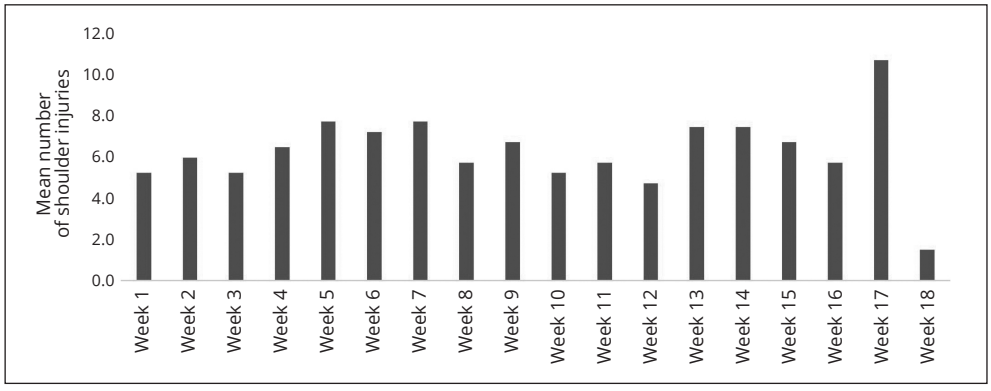

Figure 1.—Distribution of mean number of shoulder injuries in the NFL by week (seasons 2019-2022).
NFL: National Football League.

and other injuries (P=0.001). While fractures resulted in a higher number of missed games when compared to ligamentous injuries, this difference did not reach significance (P=0.118) (Table V).

Shoulder injury timing

There was no apparent shoulder injury trend when exploring its mean distribution across the football season (Figure 1). However, around 35% occurred late-season, 33.4% occurred early-season, and 31.6% occurred mid-season. Week 17 recorded the highest mean number of shoulder injuries with 10.75 injuries per season, followed by weeks 5 and 7 at 7.75 injuries per season each. The lowest number of shoulder injuries were seen during week 18, with only 1.5 injuries per season.

Discussion

The present study demonstrated how common shoulder injuries are in American professional football, with notable differences between athletes. Linebackers, cornerbacks, and wide receivers sustained more shoulder injuries than other positions, which were mostly muscular/tendinous or ligamentous in nature, often categorized as mild injuries. With regards to timing, there was no apparent shoulder injury trend throughout the football season. This study reported 452 shoulder injuries occurring over 4 seasons in the NFL, comprising 6.7% of all total injuries reported, with an injury rate of 4.52 per 1000 AE. A similar study looking at the difference of injuries in NFL before and after COVID-19, reported an upper limb injury rate of 3.06 per 1000 AE in 2018-2020 and a rate of 3.83 per 1000 AE in 2021.[16] The difference in reported rates between the aforementioned study and ours may be related to the different time periods studied, differences in methodological approaches, rule changes going into effect between seasons, and their inclusion of the playoffs following the regular season.[16] Regardless, shoulder injuries constitute a common and challenging problem for football teams and athletes in the NFL given the resulting number of games missed each season. Unsurprisingly given the full-contact nature of the sport, rates of shoulder injuries in the NFL were higher than those reported in other overhead sports like basketball, where a previous study showed an incidence rate of 1.11 injuries per 1000 AE, and higher than that reported in baseball, where a previous study reported a mean injury rate of 0.67 per 1000 AE.[22, 23] That can be explained by the fact that even though these sports share similar features in that they are highly dependent on the shoulder, the aggressive nature of football can render it more susceptible to contact injuries that can increase injury risk on the shoulder.[7] In our study, NFL players with shoulder injuries had an average age of 26 years old. This falls in accordance with the average age of NFL players, which has been suggested to vary between 25 and 27 years old, according to a news media outlet.[24] Similarly, NFL players with shoulder injuries had an average height of 188.1 cm, and an average weight of 110.5 kg. This also resembles closely the demographic characteristics of NFL players in general, who have been reported to be on average 188.4 cm tall and weigh 110.9 kg,[25] and as such, we see no significant relationships between age, height, weight and shoulder injury risk in the NFL. Offensive and defensive players similarly experienced a high number of injuries. Motions common to each position often place great stress on the shoulder joint and can lead to significant injuries.[26] Mechanisms can be divided into two categories: acute traumatic injuries and repetitive attritional/overuse injuries. Overuse injuries can occur due to repetitive movements that stress the joint, like overhead throwing or blocking.[8] The overhead throwing motion by quarterbacks

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in football is somewhat analogous to the pitching motion in baseball, which can often lead to muscular/tendinous injuries to the static and dynamic stabilizers and muscles of the shoulder, like the rotator cuff.[27] Shoulder pathologies seen in linemen can also be chronic attritional injuries leading to posterior instability. Acute traumatic injuries, on the other hand, can occur due to falls, collisions, and tackles in football, and these can lead to fractures, instability, ligamentous injuries, and rotator cuff injuries as well.[8] This variety in injury mechanism translates into a variety of shoulder injury types that can target different playing positions in specific ways. The majority of shoulder injuries in the NFL were fortunately mild injuries, on average causing the player to miss two games. Fractures and ligamentous injuries led to a greater number of missed games than muscular/tendinous and other injuries, potentially because these more severe injuries have a higher probability of requiring surgery, which can lead to extended rehabilitation and recovery times. On the other hand, muscular/tendinous injuries can often present as mild strains and soreness, many of which can be treated conservatively with rest and expectant management.[28] The average salary for an NFL player in 2022 was 2.8 million USD, and players are often paid whether or not they are able to play a game due to injury.[29] As such, and by dividing the player's salary by the number of games per season, the average shoulder injury can lead to around 329,000 USD in time loss due to injury, excluding additional medical fees. Hence, shoulder injuries can constitute an economic burden to NFL clubs and teams alike. Shoulder injuries were appropriately spread throughout the NFL seasons with reported injuries occurring from week 1 till week 18. This finding does not align with that of other sports like baseball or soccer, whose injury rates are usually higher early on in the season, and decrease with time.[21, 22, 30] The high rise in injury rates early on for these sports has been attributed to the abrupt change of pace in training and workload for both soccer and baseball following their off-season, along with the condensed competition schedule that is seen in soccer in the earlier months of the season.[21, 22, 30] The fact that such a trend was not seen in our study may be an indicator that games are appropriately spaced out in the NFL with around six days of rest between games, that the NFL preseason appropriately integrates players into the season, and that the number of players for each team is large enough to allow adequate rest and workload distribution. We did note that week 18 recorded the lowest mean number of shoulder injuries per season. This may be explained by the fact that, around that time, some teams that have already secured a first-round spot in the playoffs may be resting many of their starters who have played all season, leading to fresher players who can be potentially less susceptible to injury. Based on the results of our study, several measures can be taken by NFL governing bodies to help decrease the rates of shoulder injuries in the sport. One measure that has been discussed and implemented over the years is further expansion of roster size.[31] This can be very helpful for player rotation, decrease performance burden on individual players, and help endorse player health and safety.[31] Other measures that can be taken would be to provide appropriately fitted equipment, highlight importance of proper technique implementation, ensure gradual training loads, and emphasize training programs that aim to strengthen rotator cuff muscles.[32-34] Finally, one important measure that needs to be taken would be the establishment of a deidentified injury registry that collects detailed biomechanical and medical information regarding different injuries in the NFL. This can play a pivotal role in understanding the mechanisms behind these injuries, and subsequently establish preventative protocols and strategies.

Limitations of the study

To our knowledge, this is the first study to examine the epidemiology of shoulder injuries in the NFL, but several limitations may exist. The injury data in this study was collected through media outlets and public databases. Even though this method of data collection has been used by other studies in the literature, discrepancies in reporting may still exist, especially when compared to official NFL team records. We also only included seasons 2019-2022, as data regarding injury type for previous seasons was lacking in adequate detail. In addition, there is a tendency among NFL teams to either under-report or report mild shoulder injuries as either strain or sprain, so that opposing teams do not gain an advantage by knowing which players are prone to injury. Unfortunately, this leads to ambiguity in injury tracking and increasing the risk of reporting the wrong injury type.

Conclusions

Shoulder injuries constitute common and debilitating injuries in American football, and that these injuries constitute 6.7% of all injuries in football. Positions with a higher number of shoulder injuries were linebackers, cornerbacks and wide receivers. There was no difference in number of shoulder injuries within different defensive and specialty positions, but within offensive positions, wide receivers

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and tight ends had higher numbers of shoulder injuries when compared to centers and fullbacks. Injuries were mostly muscular/tendinous or ligamentous in nature and were typically mild injuries with an average of less than two missed games. This study provides coaches and trainers with insight into the patterns of shoulder injuries in the NFL, including positions that may benefit from more targeted training protocols to minimize risk. Clubs and medical team professionals should be encouraged to report injuries in a deidentified detailed manner to confidential health-related governing bodies, in order to better understand the mechanisms of injury in the sport and extrapolate prevention strategies and guidelines. In this setting, developing an injury registry with detailed management plans and rehabilitation protocols may hold some value.

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Conflicts of interest

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Authors' contributions

Mohamad Y. Fares and Joseph A. Abboud have given substantial contributions to the study conception; Mohamad Y. Fares, Mohammad Daher, Peter Boufadel, Jaspal Singh, and Jonathan Koa contributed to the data collection and to the manuscript first draft; Mohamad Y. Fares and Ahmad Bdeir contributed to the data analysis; Ryan Lopez, Daniel Goltz, and Joseph A. Abboud contributed to the manuscript final draft and critical revision for important intellectual content; Joseph A. Abboud contributed to the study supervision. All authors read and approved the final version of the manuscript.

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